

TECHNICAL REPORT

Computational Discovery of a Dual Neprilysin / DPP-IV Inhibitor

A structure-based virtual screening pipeline for type 2 diabetes
in patients at risk of heart failure with reduced ejection fraction

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Scope and provenance. This report documents a virtual screening methodology developed during the author's final-year undergraduate research project. The original scripts and intermediate datasets were not preserved under version control, and this document is therefore a reconstruction of the methodology rather than a record of a specific set of results. Numerical values presented in figures and tables are illustrative of how the pipeline behaves, and are labelled as such throughout. No compound described here has been synthesised or biologically assayed.

In the original project, machine learning model training and generative expansion of the scaffold libraries were carried out by a doctoral colleague rather than by the author. Section 15 explains that component in full for the benefit of readers wishing to reproduce the work, and flags this attribution where relevant.

Accompanying code and documentation:

github.com/SupremeMaxitout/nep-dpp4-dual-inhibitor-screen

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Contents

Summary	2
Part I: Scientific Rationale	3
1 The clinical problem	3
2 Neprilysin as a cardiovascular target	3
2.1 Active site architecture	3
3 DPP-IV as a metabolic target	3
3.1 Active site architecture	4
4 The case for dual inhibition, and the cautionary precedent	4
5 The central obstacle: a pharmacophore conflict	4
6 Hypothesis and objectives	5
Part II: Methodology	6
7 Pipeline overview	6
8 Stage 1: target preparation	7
8.1 Structure selection	7
8.2 Structure preparation	7
8.3 Search box definition	7
9 Stage 2: protocol validation	7
9.1 Redocking	7
9.1.1 Symmetry correction in RMSD calculation	7
9.2 Enrichment benchmarking	8
9.2.1 Why decoys must be property-matched	8
9.2.2 Metrics and their interpretation	8
9.3 Threshold calibration	10
10 Stage 3: library construction	10
10.1 Source and filtering	10
10.2 Aggregating repeated measurements	10
10.3 Chemical standardisation	10
10.4 Reporting attrition	11
11 Stage 4: primary docking and rescoring	11
11.1 Docking engines	11
11.2 Ranking by ligand efficiency	11
12 Stage 5: cross-docking and dual-activity selection	11
12.1 A methodological correction	11

13 Stage 6: scaffold analysis and enumeration	12
13.1 Scaffold extraction	12
13.2 The dual-pharmacophore constraint	12
13.3 Analogue enumeration	12
14 Stage 7: antitarget counter-screening	13
 Part III: The Machine Learning Component	 14
15 Machine learning in this pipeline	14
15.1 What problem the model is solving	14
15.2 Representing a molecule as numbers	14
15.2.1 Molecular descriptors	14
15.2.2 Fingerprints	14
15.2.3 Learned representations	14
15.3 The label problem	15
15.3.1 The data that was already available	15
15.4 Multi-task architecture	15
15.5 Choosing a model	16
15.6 Validation: the part that is most often done wrong	16
15.6.1 Why random splitting overestimates performance	16
15.6.2 Scaffold splitting	17
15.6.3 Applicability domain	17
15.7 A recommended implementation sequence	17
 Part IV: Appraisal and Reproduction	 18
16 Divergences from the original implementation	18
17 Limitations	18
18 Reproducing this work	19
18.1 Prerequisites	19
18.2 Procedure	19
18.3 Compute	19
19 Concluding remarks	20

Summary

Patients with type 2 diabetes and heart failure with reduced ejection fraction (HFrEF) currently require separate drugs for each condition. A single molecule inhibiting both neprilysin, a validated heart failure target, and dipeptidyl peptidase-IV, a validated glycaemic target, would reduce pill burden and remove a drug-drug interaction surface.

This report describes a computational pipeline designed to identify such molecules. It proceeds in six stages: preparation and validation of the target structures, construction of bioactivity-filtered compound libraries, primary docking with rescoring, cross-docking to identify compounds active at both targets, extraction and constrained enumeration of shared molecular scaffolds, and counter-screening against the antitargets whose inhibition has historically caused clinical failure in this therapeutic space.

The central scientific difficulty is stated plainly at the outset rather than discovered late. Neprilysin requires an anionic zinc-binding group; DPP-IV requires a basic nitrogen. A molecule satisfying both is a zwitterion, with the permeability penalty that entails. The pipeline is designed around this constraint rather than in ignorance of it.

The report is written so that a reader with a background in chemistry or biology, but no prior knowledge of this project, can follow the reasoning and reproduce the work independently. Where the methodology has been improved relative to what was originally done, both versions are described and the reasons given.

Part I Scientific Rationale

1 The clinical problem

Type 2 diabetes and heart failure occur together far more often than chance would predict. Diabetes accelerates cardiac remodelling, and the failing heart in turn worsens insulin resistance. The consequence is a large population of patients carrying both diagnoses, each managed with its own drug class, its own titration schedule and its own adverse event profile.

Within heart failure, the subgroup of interest here is heart failure with reduced ejection fraction, in which the left ventricle contracts inadequately. This subgroup is where pharmacological intervention has the strongest evidence base, and where neprilysin inhibition has been definitively established.

Polypharmacy in this population is not a cosmetic problem. Adherence falls as the number of daily tablets rises, and every additional agent expands the interaction matrix a prescriber must reason about. A single molecule addressing both conditions is therefore of genuine clinical interest, provided it can be made to work.

2 Neprilysin as a cardiovascular target

Neprilysin (NEP, neutral endopeptidase, EC 3.4.24.11, CD10) is a membrane-bound zinc metallopeptidase. Among its substrates are the natriuretic peptides, which promote vasodilation, natriuresis and diuresis. Inhibiting neprilysin raises circulating natriuretic peptide levels, which unloads the failing heart.

Neprilysin inhibition alone proved insufficient in clinical testing. Combined with angiotensin receptor blockade, however, it produced the drug sacubitril/valsartan, which demonstrated a clear mortality benefit over enalapril in HFrEF and is now standard of care. Sacubitril is a prodrug; its active metabolite, LBQ657 (sacubitrilat), is the species that binds neprilysin.

2.1 Active site architecture

The catalytic machinery sits inside a large internal cavity, an arrangement that classifies neprilysin as a cryptidase. The catalytic zinc ion is coordinated by the side chains of His583, His587 and Glu646, with the fourth coordination position available to an incoming ligand. In the neprilysin/LBQ657 complex, that fourth position is occupied by a carboxylate oxygen of the inhibitor at a distance of approximately 2.0 Å, with the second carboxylate oxygen a further 2.7 Å away [4].

The single most important fact for this project. Potent neprilysin inhibition is driven by direct coordination of the catalytic zinc by an anionic group on the ligand. Carboxylates, thiols, hydroxamates and phosphinates all serve this role. A neprilysin inhibitor lacking such a group is, with rare exceptions, not a neprilysin inhibitor.

This fact has a direct computational consequence, discussed in Section 11.1: the scoring function used for docking must be able to represent metal coordination, and the most widely used one cannot.

3 DPP-IV as a metabolic target

Dipeptidyl peptidase-IV (DPP-4, CD26) is a serine protease that cleaves and thereby inactivates the incretin hormones GLP-1 and GIP. Inhibiting it prolongs incretin signalling, which enhances glucose-dependent insulin secretion. The resulting drug class, the gliptins, is widely used and generally well tolerated.

3.1 Active site architecture

The enzyme comprises an eight-bladed beta-propeller domain and an alpha/beta hydrolase domain carrying the catalytic triad Ser630, Asp708 and His740. Two subsites dominate inhibitor design. The S1 pocket is small and hydrophobic, lined by Tyr631, Val656, Trp659, Tyr662, Tyr666 and Val711. The S2 subsite contains the glutamate dyad Glu205 and Glu206.

In the sitagliptin complex (PDB 1X70), the inhibitor's primary amine forms hydrogen bonds and salt bridge interactions with Glu205, Glu206 and Tyr662, while the trifluorophenyl group occupies S1 [5].

The corresponding fact for DPP-IV. The Glu205/Glu206 dyad anchors a basic nitrogen, and essentially every rationally designed DPP-IV inhibitor carries a primary or secondary amine for this purpose.

4 The case for dual inhibition, and the cautionary precedent

Combining the two mechanisms in one molecule is attractive for the reasons given in Section 1. It is also a strategy with a specific and instructive history of failure.

Omapatrilat was a vasopeptidase inhibitor designed to inhibit both neprilysin and angiotensin-converting enzyme (ACE). Both enzymes degrade bradykinin, and inhibiting both simultaneously raised bradykinin levels enough to cause angioedema at clinically unacceptable rates. In the OCTAVE hypertension trial, angioedema occurred in 2.17% of omapatrilat-treated patients compared with 0.68% on enalapril, with markedly higher rates in Black patients [2]. In the OVERTURE heart failure trial, omapatrilat was non-inferior but not superior to enalapril, and angioedema was again more frequent [1]. The drug was not approved.

A parallel lesson exists on the DPP-IV side. DPP-8 and DPP-9 share catalytic machinery with DPP-IV, and inhibitors lacking selectivity over them produced alopecia, thrombocytopenia, reticulocytopenia, splenomegaly, multi-organ histopathological change and mortality in rodents, together with gastrointestinal toxicity in dogs [3]. Selectivity over DPP-8/9 has been a standard requirement for the class ever since.

Design consequence. Any programme combining neprilysin inhibition with a second mechanism must demonstrate selectivity over ACE, and any programme involving DPP-IV must demonstrate selectivity over DPP-8/9. In this pipeline these are not optional late-stage checks. They are built in as an explicit counter-screening stage (Section 14).

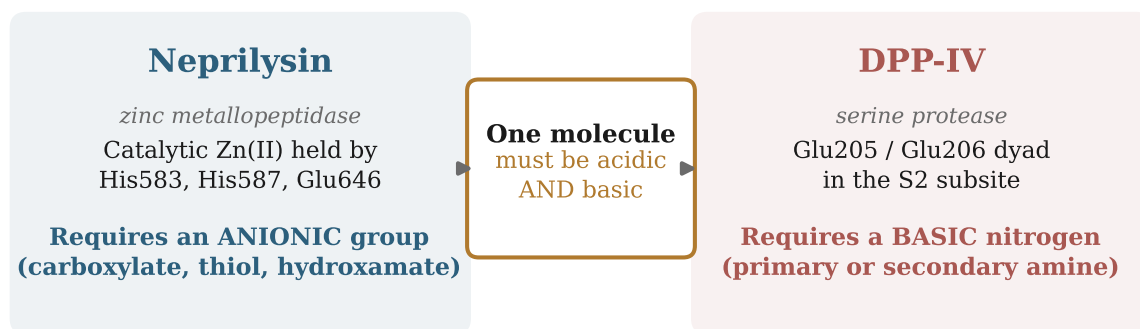
5 The central obstacle: a pharmacophore conflict

Sections 2 and 3 establish two requirements that sit awkwardly together. NEP demands an anionic group. DPP-IV demands a basic nitrogen. A molecule carrying both will, at physiological pH, exist substantially as a zwitterion.

Zwitterions are poorly membrane-permeant. The practical evidence sits in the clinic already: sacubitril is dosed as an ethyl ester prodrug precisely because its active carboxylate form does not adequately cross membranes.

This does not make the project impossible. It makes the project honest about what it is looking for. Three outcomes are plausible.

1. A zwitterionic molecule with adequate permeability, most likely small and conformationally constrained, possibly exploiting an intramolecular hydrogen bond to mask polarity.
2. A prodrug strategy, masking the acid as an ester and relying on esterase-mediated activation, following the sacubitril precedent.
3. A molecule using a neutral zinc-binding group such as a hydroxamate, trading some neprilysin potency for improved physical properties.



Consequence: the molecule is a zwitterion at physiological pH.

High polarity and low membrane permeability follow directly.

Sacubitril is administered as an ester prodrug for exactly this reason.

Figure 1: The dual pharmacophore requirement. The two targets demand opposing ionisable groups. A molecule satisfying both is necessarily zwitterionic, which carries a direct permeability cost. This is a chemical constraint the pipeline must work within rather than a problem it can solve.

A screening campaign that does not state this constraint up front will discover it expensively, several months in, when the surviving compounds all turn out to be undevelopable.

6 Hypothesis and objectives

Hypothesis. Within the space of compounds with reported bioactivity against either neprilysin or DPP-IV, there exist molecular scaffolds capable of satisfying the binding requirements of both targets simultaneously. Such scaffolds can be identified computationally by cross-docking, isolated by scaffold analysis, and elaborated into a focused set of candidates suitable for experimental testing.

The objectives follow directly:

1. Establish and validate a docking protocol for each target, with demonstrated ability to discriminate actives from decoys rather than merely to reproduce a crystal pose.
2. Assemble curated compound libraries for each target from public bioactivity data.
3. Identify compounds predicted to bind both targets.
4. Extract the molecular scaffolds common to those compounds.
5. Enumerate analogues under an explicit dual-pharmacophore constraint.
6. Eliminate compounds predicted to engage ACE or DPP-8/9.
7. Deliver a prioritised, selectivity-filtered candidate set.

Part II Methodology

7 Pipeline overview



Figure 2: The complete workflow. Each stage produces a human-readable table and a provenance record. Stages 1 and 4 act as gates: the pipeline should not proceed past a failed validation or an empty dual-active set.

The design principle throughout is that every stage must produce an inspectable intermediate and every filter must be justifiable. A virtual screen that cannot explain why a particular compound survived is not a screen but a lottery with extra steps.

8 Stage 1: target preparation

8.1 Structure selection

Suitable structures meet five criteria: human protein; a co-crystallised inhibitor present (holo rather than apo); resolution 2.5 Å or better; no missing residues or disordered loops within the binding site; and, for neprilysin, the catalytic zinc present in the deposited coordinates.

The requirement for a bound ligand is not incidental. The co-crystallised ligand serves two distinct purposes downstream: it defines the centre of the docking search box, and it provides the reference pose against which the protocol is validated. An apo structure provides neither, and its side chains frequently adopt conformations that no ligand can occupy.

8.2 Structure preparation

Preparation removes waters and crystallisation additives, retains one protein chain, and converts the result to the PDBQT format required by AutoDock.

A failure mode worth naming explicitly. A routine cleanup step of the form “remove all HETATM records” deletes the catalytic zinc along with the buffer molecules. Every subsequent calculation then proceeds normally and produces plausible-looking output that is entirely wrong, because the single most important feature of the neprilysin binding site is no longer present. The pipeline maintains a whitelist of functional metal ions, and verifies after format conversion that the metal survived. A silent failure is converted into a loud one.

8.3 Search box definition

The box centre is the centroid of the co-crystallised ligand. The box dimensions are taken from configuration, but are enlarged automatically if the ligand’s own bounding box plus 4 Å of padding would exceed them. A box smaller than the reference ligand cannot reproduce the crystallographic pose, which would render the validation in Stage 2 meaningless.

9 Stage 2: protocol validation

This stage exists to answer a question that is easy to skip and expensive to skip: does this docking protocol actually work for this target?

9.1 Redocking

The co-crystallised ligand is removed, prepared independently, and docked back into its own site. If the top-ranked pose reproduces the crystallographic pose to within 2.0 Å root-mean-square deviation, the protocol can at least reproduce a known answer.

9.1.1 Symmetry correction in RMSD calculation

Comparing two poses atom by atom is incorrect whenever the ligand possesses topological symmetry. The two oxygen atoms of a carboxylate are chemically indistinguishable; a pose in which they are exchanged is the same pose. A naive index-based RMSD nevertheless scores such a pose as substantially displaced.

The pipeline uses a symmetry-corrected RMSD that enumerates substructure matches and returns the best alignment, and reports the naive value alongside so that the magnitude of the correction is visible.

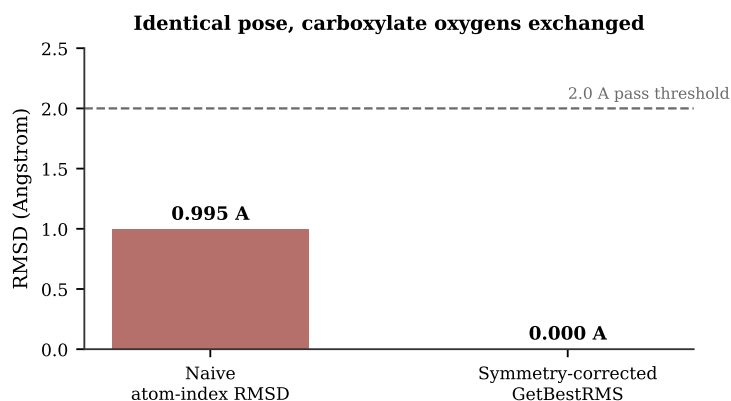


Figure 3: Measured on a phenylacetic acid in which only the two carboxylate oxygens are exchanged, leaving the pose chemically identical. Naive atom-index RMSD reports 0.995 Å of error that does not exist. Symmetry-corrected RMSD correctly reports zero. For the carboxylate-bearing chemotypes central to neprilysin inhibition, this difference can determine whether a validation is reported as a pass or a failure.

9.2 Enrichment benchmarking

Redocking demonstrates that a protocol can reproduce a geometry to which it was, in effect, given the answer. It says nothing about whether the protocol can rank an active compound above an inactive one, which is the only thing a virtual screen actually does.

The enrichment benchmark addresses this. Known actives are mixed with decoys, the mixture is docked, and the extent to which actives rise to the top of the ranking is measured.

9.2.1 Why decoys must be property-matched

If decoys were drawn at random, they would differ from the actives in molecular weight, charge and lipophilicity. Docking would then separate the two sets on those trivial grounds, producing excellent apparent enrichment that reflects nothing about binding. This is the construction principle behind the DUD-E benchmark set [9].

Each decoy is therefore matched to an active within tolerance on six properties, while being constrained to remain topologically dissimilar so that a genuine active cannot enter the decoy set by accident.

Table 1: Decoy matching criteria. The property tolerances remove the trivial discriminators; the similarity ceiling prevents true actives contaminating the decoy set.

Criterion	Tolerance	Purpose
Molecular weight	± 25 Da	remove size as a discriminator
Calculated logP	± 1.0	remove lipophilicity as a discriminator
H-bond donors	± 1	match polarity profile
H-bond acceptors	± 2	match polarity profile
Rotatable bonds	± 2	match flexibility
Formal charge	exact	charge state strongly affects docking
Tanimoto similarity	≤ 0.35	exclude probable true actives

9.2.2 Metrics and their interpretation

Two further points deserve emphasis. First, enrichment factor at 1% matters more in practice than ROC-AUC, because compounds are only ever purchased or synthesised from the top of the ranked list; performance in the tail is irrelevant. Second, BEDROC with $\alpha = 20$ weights approximately the top 8% of the list and is reported for the same reason [10].

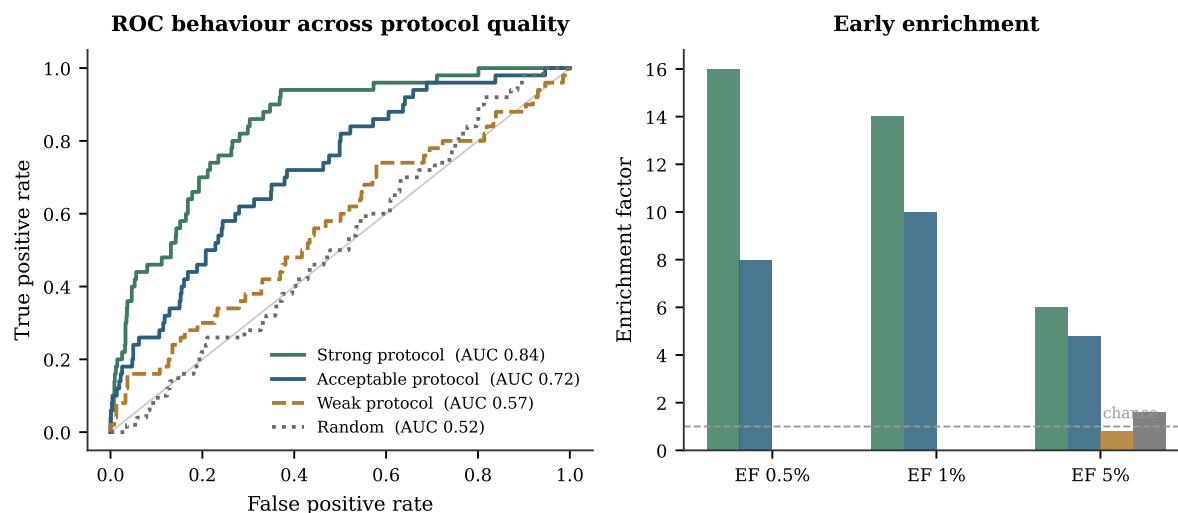


Figure 4: Illustrative benchmark behaviour across four protocol qualities, computed on simulated score distributions with 50 actives among 1000 compounds. Left: receiver operating characteristic curves. Right: enrichment factors at three list depths. Note that enrichment factor has a ceiling set by the total-to-active ratio, here 20. These values demonstrate how the metrics respond and are not measurements from this project.

Table 2: Interpreting the enrichment benchmark.

ROC-AUC	Verdict	Action
below 0.60	Unusable	The protocol ranks little better than chance. Screening with it is an expensive random draw. Revisit preparation, box definition and metal handling.
0.60 to 0.70	Weak	Usable for coarse triage only. Do not trust the ranking within the surviving set.
0.70 to 0.85	Acceptable	A normal docking result for a well-behaved target. Proceed.
above 0.85	Verify first	Unusually strong. Check for a property leak in the decoy construction before accepting it.

9.3 Threshold calibration

The original project applied a fixed selection threshold of -9.0 kcal/mol. Such a number is typically inherited from a previous publication and carries no guarantee of appropriateness for a different target or protocol.

This pipeline instead derives the threshold from the benchmark: the score capturing the desired percentile of the ranked list is computed and written to the configuration for downstream stages. A cutoff derived from the protocol's own measured behaviour is defensible in a way that a borrowed constant is not.

10 Stage 3: library construction

10.1 Source and filtering

Compounds are retrieved from ChEMBL [11], which aggregates bioactivity measurements from the primary literature. The aggregation is valuable and the variation in data quality is considerable, so four filters are applied at the point of retrieval.

Table 3: Bioactivity filters applied during retrieval. Each discards data deliberately.

Filter	Setting	Rationale
Assay type	Binding (B)	Functional and ADMET readouts measure something else
Confidence score	≥ 8	ChEMBL's own rating of target assignment certainty
pChEMBL value	must be present	Guarantees a comparable normalised number
Standard relation	equals	Excludes inequalities such as "greater than 10 μ M"

10.2 Aggregating repeated measurements

A single compound is frequently measured in several independent studies. These are collapsed to the median, which resists the single extreme outlier that a mean would follow.

The range across measurements is retained as a separate field. In testing, a compound reported at pChEMBL 4.5 in one assay and 9.0 in another was correctly assigned the median of 6.75 and flagged with a spread of 4.5 log units. A disagreement of that magnitude means one of the two reports is wrong, and ChEMBL alone cannot establish which. Such compounds should be treated as uncertain rather than silently averaged.

10.3 Chemical standardisation

The cleaning sequence is fixed and the order is load-bearing:

parse $\rightarrow$ strip salts $\rightarrow$ standardise $\rightarrow$ reject PAINS $\rightarrow$ compute properties $\rightarrow$ property filter $\rightarrow$ deduplicate

Why deduplication comes last. If compounds are deduplicated before salts are stripped, a hydrochloride salt and its free base are recorded as two distinct entries and both survive into the library. Stripping first collapses them to a single compound. This was verified directly: a test set containing both forms of the same molecule correctly resolved to one entry.

Why charges are preserved. Standard cheminformatics practice neutralises formal charges during standardisation. That practice is wrong for this project. The dual pharmacophore of Section 5 is zwitterionic by construction, and neutralising charges would erase precisely the feature being selected for.

PAINS substructures, which produce apparent activity across unrelated assays through mechanisms unrelated to specific binding, are rejected [14].

Physicochemical limits are deliberately looser than the rule of five. A strict rule-of-five gate would discard the zwitterions this project is looking for. The filters exclude the clearly undevelopable without prejudging the chemistry.

10.4 Reporting attrition

The library builder reports why compounds were rejected, not merely how many survived. If a large fraction fail on a single property, that property window is wrong, and the investigator needs to know before committing compute to docking.

11 Stage 4: primary docking and rescoring

11.1 Docking engines

AutoDock Vina [6, 7] is used for DPP-IV, where it performs well. It is not used for neprilysin, and the reason is central.

Vina cannot score zinc coordination. Its scoring function contains no dedicated term for metal coordination. The catalytic zinc is treated as a generic atom with van der Waals and electrostatic character, with no representation of the directional tetrahedral geometry that governs coordination chemistry.

Since neprilysin inhibition is driven almost entirely by zinc chelation (Section 2.1), a Vina ranking for this target systematically mis-scores the single interaction that determines potency.

Neprilysin is therefore docked with AutoDock4 using the AD4Zn force field [8], which places four tetrahedrally arranged pseudo-atoms around the zinc to reproduce its directional coordination preference.

11.2 Ranking by ligand efficiency

Raw docking scores scale with molecular size, because a larger molecule makes more contacts. A ranking by score alone is therefore partly a ranking by molecular weight, and tends to promote large, lipophilic compounds that are poor starting points for optimisation.

Ligand efficiency normalises for this:

$$LE = \frac{-\Delta G_{\text{bind}}}{N_{\text{heavy atoms}}} \quad (1)$$

Both rankings are computed and reported. Where they disagree substantially, that disagreement is itself informative about the composition of the hit set.

12 Stage 5: cross-docking and dual-activity selection

Compounds surviving their own target are docked against the opposite target. Those passing threshold on both constitute the dual-active set.

12.1 A methodological correction

The original project bridged the two libraries differently. Each compound in the neprilysin hit set was compared by two-dimensional Tanimoto similarity against each compound in the DPP-IV hit set, and only sufficiently similar pairs were then cross-docked.

This was a reasonable heuristic for saving compute, and it embeds an assumption that does not hold: that two-dimensional fingerprint similarity implies a shared binding mode. It does not. Two molecules can be highly similar by fingerprint and bind entirely differently, or be dissimilar and share a pharmacophore.

Cross-docking the full hit sets in both directions answers the question directly and removes the assumption. The computational cost is real but modest at the scale involved, and it is now affordable in a way it was not previously.

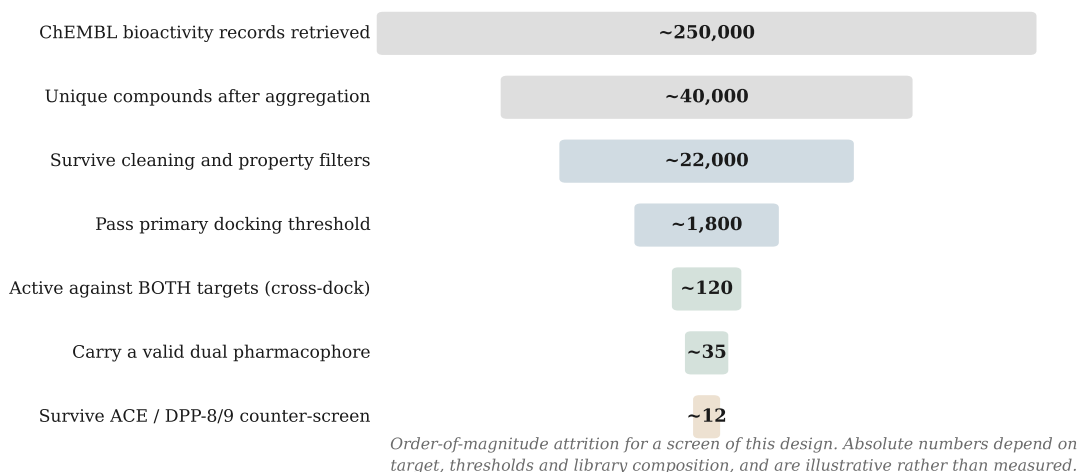


Figure 5: Attrition through the pipeline. The compound count falls by roughly four orders of magnitude from initial retrieval to the final prioritised set. The sharpest single drop occurs at the dual-activity intersection, which is the expected consequence of the pharmacophore conflict described in Section 5.

13 Stage 6: scaffold analysis and enumeration

13.1 Scaffold extraction

Molecular scaffolds are extracted using the Bemis and Murcko definition [13]: ring systems plus the linkers joining them, with side chains removed. Scaffolds are ranked by how frequently they recur across the dual-active set.

In the original project, the scaffolds carried forward were selected manually by the project supervisor. Expert selection is legitimate, but it is not reproducible by a third party. This pipeline ranks scaffolds by explicit criteria, which makes the selection auditable.

13.2 The dual-pharmacophore constraint

This is the point at which the analysis of Section 5 becomes an executable filter rather than a background observation. Scaffolds must carry both required features, expressed as SMARTS substructure patterns:

Making the requirement explicit converts a hope into a specification. It also makes subsequent failures interpretable: if no scaffold satisfies both patterns, that is a meaningful negative result about the accessible chemical space, not an unexplained empty output.

13.3 Analogue enumeration

Selected scaffolds are elaborated into analogue sets. The original project used DataWarrior for this, which works but introduces a graphical step into an otherwise scriptable pipeline and therefore breaks end-to-end reproducibility. Enumeration here is performed programmatically.

Table 4: The dual-pharmacophore constraint, expressed as substructure queries. A scaffold must match at least one pattern from each group.

Target	Feature	SMARTS
NEP	carboxylate	[CX3] (=O) [OX1H0- , OX2H1]
NEP	hydroxamate	[CX3] (=O) [NX3] [OX2H1]
NEP	thiol	[SX2H]
NEP	phosphinate	[PX4] (=O) ([OX1H0- , OX2H1])
DPP-IV	primary amine	[NX3 ; H2 ; !\$(NC=O)]
DPP-IV	secondary amine	[NX3 ; H1 ; !\$(NC=O) ; !\$(N-a)]

Enumerated structures are filtered on synthetic accessibility [15]. Unconstrained enumeration generates large numbers of structures that are formally valid and practically unmakeable, and filtering them early avoids wasting docking capacity on molecules no chemist could deliver.

14 Stage 7: antitarget counter-screening

Surviving compounds are docked against the antitargets identified in Section 4.

Table 5: Antitarget panel and the clinical evidence motivating each.

Antitarget	Relevance	Evidence
ACE	NEP-linked	Dual NEP/ACE inhibition by omapatrilat caused angioedema at 2.17% versus 0.68% for enalapril in OCTAVE, preventing approval [2, 1].
DPP-8 / DPP-9	DPP-IV-linked	Non-selective inhibition produced alopecia, thrombocytopenia, reticulocytopenia, splenomegaly, multi-organ pathology and mortality in rodents, and gastrointestinal toxicity in dogs [3].

Compounds scoring well against an antitarget are penalised rather than discarded outright, since the selectivity margin is itself a property to be optimised rather than a binary gate.

Part III The Machine Learning Component

15 Machine learning in this pipeline

Attribution. In the original project, this component was implemented by a doctoral colleague rather than by the present author. It is described here in full so that the pipeline can be reproduced end to end, and because the methodological issues it raises are instructive and widely shared.

15.1 What problem the model is solving

By the end of Stage 6, enumeration has produced tens of thousands of candidate structures. Docking all of them is feasible but slow, and docking is itself only an approximation. The purpose of the model is triage: to learn the relationship between molecular structure and predicted activity from the compounds already scored, and then to prioritise the remainder without docking each one.

This is a supervised regression problem. The input is a molecule. The output is a number representing predicted activity. The model learns the mapping from examples.

15.2 Representing a molecule as numbers

A machine learning model cannot read a chemical structure directly. The molecule must be converted into a fixed-length numeric vector, a process called featurisation. Three approaches are common.

15.2.1 Molecular descriptors

The simplest option. Compute a set of interpretable numbers for each molecule: molecular weight, calculated logP, topological polar surface area, counts of hydrogen bond donors and acceptors, rotatable bonds, ring counts, and so on. A few hundred such descriptors are readily available.

Strengths: interpretable, cheap, robust with small datasets. Weakness: two molecules with very different structures can share almost identical descriptor profiles, so the representation discards structural information the model may need.

15.2.2 Fingerprints

The standard approach in this domain. An extended-connectivity fingerprint (ECFP) examines the local environment of every atom out to a fixed bond radius, hashes each environment into an integer, and sets the corresponding bit in a fixed-length vector [12]. ECFP4, meaning radius 2, in 2048 bits is the usual default.

The result is a binary vector in which each position indicates the presence or absence of a particular substructural motif. Similar molecules share many bits. This representation is cheap, deterministic, and empirically hard to beat on datasets of the size typically available in a project like this one.

15.2.3 Learned representations

Graph neural networks treat the molecule as a graph of atoms and bonds and learn the featurisation jointly with the prediction task. This is the modern approach and can outperform fingerprints, but it requires substantially more data to train well. On a few thousand compounds it typically does not beat a well-tuned gradient boosting model on ECFP4.

Practical recommendation. Start with ECFP4 concatenated with a handful of physicochemical descriptors. It is a strong baseline that is difficult to beat on datasets below roughly ten thousand compounds, and it establishes whether a more complex model is earning its complexity.

15.3 The label problem

This is the most important section in Part III, and the point on which the original project can be most usefully criticised.

A supervised model learns to predict whatever it is trained on. The choice of label therefore determines what the model actually learns, and no amount of architectural sophistication can compensate for the wrong choice.

The original model was trained on docking scores. The reasoning was practical: docking scores were available for every enumerated compound, and experimental data was not available for most of them.

The consequence. A model trained on docking scores learns to approximate the scoring function, not the biology. Its accuracy ceiling is the accuracy of the docking programme it was trained against. A model that reproduced Vina perfectly would inherit every one of Vina's errors, including the systematic mis-scoring of zinc coordination described in Section 11.1.

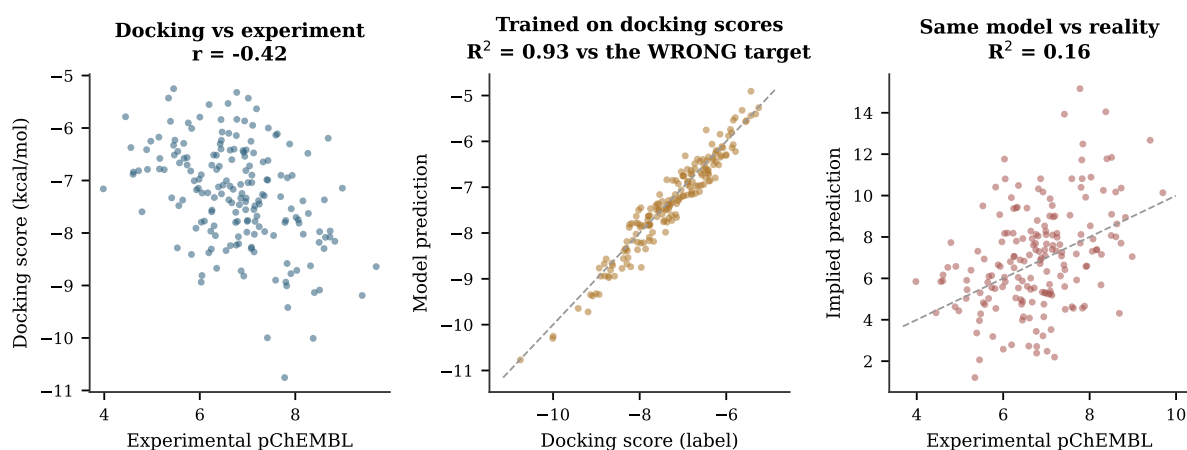


Figure 6: Why the label choice dominates. Left: docking scores correlate only weakly with experimental potency, which is a well-documented property of docking rather than a defect of any particular implementation. Centre: a model trained on docking scores can reproduce them with high fidelity. Right: that same model, evaluated against experimental potency, performs poorly. The model is excellent at the task it was given and the task was the wrong one. Simulated data illustrating the structure of the problem.

15.3.1 The data that was already available

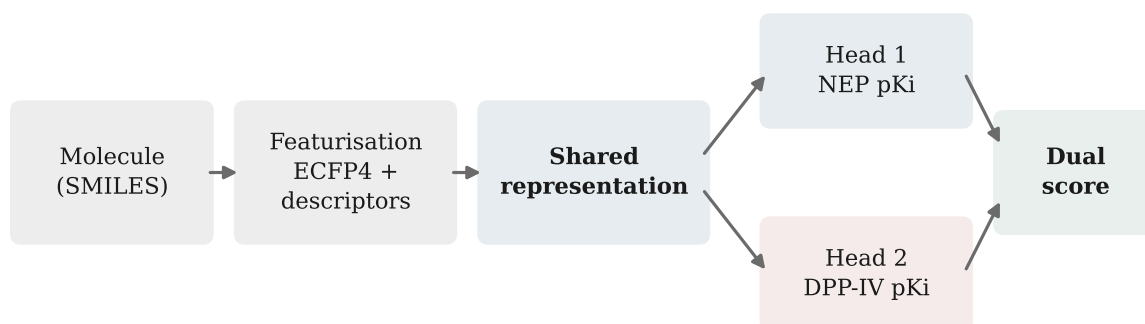
The better approach required no additional experiments and no additional compute. Stage 3 of this pipeline mines ChEMBL for compounds *filtered by their measured IC₅₀ values*. Those experimental potency labels were downloaded in order to apply the filter, and were then discarded in favour of computed scores.

A model trained on experimental pChEMBL values for both targets would have used data that was already in hand. The docking scores could still contribute, as an auxiliary prediction task or as a pre-training signal, without being treated as ground truth.

15.4 Multi-task architecture

Because the project needs predictions for two targets, the natural design shares most of the model between them.

The reasoning is that binding to either target depends on overlapping chemical properties: shape, lipophilicity, hydrogen bonding capacity, charge distribution. A shared layer can learn these once from the combined data of both targets, which is particularly valuable when one target has substantially fewer measurements than the other.



The shared layer learns chemistry relevant to both targets. Each head specialises. Training on experimental pKi rather than docking scores keeps the labels tied to biology.

Figure 7: Multi-task architecture. A shared representation learns chemistry relevant to both targets, while separate output heads specialise. Training on experimental potency keeps the labels tied to biology rather than to a scoring function.

15.5 Choosing a model

Table 6: Model selection for datasets of the size typical in this setting.

Model	When to use	Notes
Random Forest	First baseline	Almost impossible to configure badly. If a more complex model cannot beat it, the complexity is not earning its place.
Gradient boosting (XGBoost, LightGBM)	Default choice	Usually the best performer on fingerprint input at this scale [16]. Handles sparse binary features well.
Neural network (MLP)	Multi-task settings	The natural choice when a shared representation with multiple heads is wanted.
Graph neural network	Large datasets	Requires more data than this kind of project usually has. Worth revisiting above roughly ten thousand measured compounds.

15.6 Validation: the part that is most often done wrong

A model that performs well on data resembling its training set may perform very poorly on genuinely novel chemistry. Since the purpose here is to prioritise *new* compounds, the evaluation must reflect that.

15.6.1 Why random splitting overestimates performance

Chemical datasets contain congeneric series: groups of closely related analogues from the same medicinal chemistry programme. A random train/test split places some members of a series in training and others in test. The model then predicts a compound nearly identical to one it has already seen, and appears highly accurate.

15.6.2 Scaffold splitting

The correct approach for this application is to split by scaffold, so that all compounds sharing a Bemis-Murcko scaffold fall on the same side of the split. The test set then contains chemotypes the model has genuinely not encountered, which is the situation it will face in deployment.

Expect the number to fall. A model scoring $R^2 = 0.85$ under random splitting may score 0.45 under scaffold splitting. The second number is the honest one. Report it. A reviewer who sees only the optimistic figure and later discovers the splitting strategy will discount everything else in the work.

15.6.3 Applicability domain

A model asked to predict a molecule unlike anything in its training set will still return a number, and that number will be unreliable. Quantifying the distance from each prediction to the training distribution, for example through maximum Tanimoto similarity to any training compound, allows low-confidence predictions to be flagged rather than silently trusted.

15.7 A recommended implementation sequence

1. Assemble experimental pChEMBL values for both targets from Stage 3 output.
2. Featurise with ECFP4 (2048 bits) plus a small descriptor set.
3. Split by Bemis-Murcko scaffold, approximately 80/20.
4. Train a Random Forest as a baseline and record its scaffold-split performance.
5. Train a multi-task gradient boosting or neural model with one head per target.
6. Compare honestly. If the complex model does not beat the baseline, use the baseline.
7. Compute applicability domain distances for every prediction.
8. Apply the model to enumerated compounds, and dock only the top-ranked fraction.
9. Report both split strategies, making clear which is which.

Part IV Appraisal and Reproduction

16 Divergences from the original implementation

This report describes an improved pipeline. The table below records every point at which it departs from what was originally done, so that the two are never confused.

Table 7: Changes relative to the original undergraduate implementation.

Stage	Originally	Now, and why
Validation	Redocking with visual pose inspection	Redocking retained, plus enrichment benchmarking against property-matched decoys. Geometric reproduction does not demonstrate the ability to discriminate actives from inactives.
NEP docking	AutoDock Vina	AutoDock4 with AD4Zn. Vina has no metal coordination term, and neprilysin inhibition is driven by zinc chelation.
File preparation	AutoDockTools / MGLTools (Python 2)	Maintained Python 3 equivalents. Output is comparable; the legacy toolchain is effectively unsupported.
Threshold	Fixed at -9.0 kcal/mol	Calibrated from the enrichment benchmark, with a parallel ligand efficiency ranking to remove the size bias.
Library bridge	2D Tanimoto similarity, then cross-docking of similar pairs	Direct cross-docking of both full hit sets. Fingerprint similarity does not imply a shared binding mode.
Scaffold choice	Manual selection by the supervisor	Frequency-ranked extraction with an explicit dual-pharmacophore SMARTS filter. Reproducible by a third party.
Enumeration	DataWarrior (graphical)	Programmatic enumeration with synthetic accessibility filtering. Keeps the pipeline scriptable end to end.
Antitargets	Not screened	ACE and DPP-8/9 counter-screens added. Given the omapatrilat precedent these are not optional.
ML labels	Docking scores	Experimental pChEMBL values recommended, with docking as an auxiliary signal. See Section 15.

17 Limitations

Stated plainly, because a virtual screen that does not state its limitations should not be trusted.

1. **No experimental validation.** No compound described here has been synthesised or assayed. Everything is prediction.
2. **Docking scores are weak predictors of affinity.** The correlation between docking score and measured binding is modest. Scores are used here for triage and ranking, not as quantitative predictions.

3. **Receptor flexibility is not modelled.** Docking treats the protein as rigid. Both targets are known to undergo induced-fit rearrangement on ligand binding, including a concerted conformational change at Trp693 and Phe106 in neprilysin when a bulky group occupies the S1' subsite [4].
4. **The zwitterion problem is surfaced, not solved.** The pipeline identifies molecules satisfying both pharmacophores. It does not make them permeable.
5. **Structure-based screening finds what resembles what is known.** Libraries are built from compounds with reported activity, which biases the search toward established chemotypes.
6. **Absolute numbers in this report are illustrative.** Figures demonstrating metric behaviour use simulated data, and this is stated in every relevant caption.

18 Reproducing this work

18.1 Prerequisites

Table 8: Software requirements.

Component	Purpose	Availability
RDKit	Cheminformatics throughout	conda-forge, open source
AutoDock Vina 1.2	Docking (DPP-IV)	conda-forge, open source
AutoDock4 + AD4Zn	Docking (NEP, zinc-aware)	Scripps, free
Open Babel	Format conversion	conda-forge, open source
ADFR Suite	Receptor preparation	Scripps, free
scikit-learn	Metrics and baseline models	conda-forge, open source
ChEMBL	Bioactivity source	Public REST API

18.2 Procedure

1. Select and verify target structures against the criteria in Section 8.1. Record the accession codes, chains and ligand codes chosen, and the reasons.
2. Prepare the receptors, retaining the catalytic zinc. Confirm it survived format conversion.
3. Validate. Redock the reference ligands, using symmetry-corrected RMSD. Then build a property-matched decoy set and measure ROC-AUC and enrichment. Do not proceed past a failing validation.
4. Build the libraries from ChEMBL with the filters in Section 10, and inspect the attrition report before continuing.
5. Dock each library against its own target. Rank by affinity and by ligand efficiency.
6. Cross-dock in both directions. Take the intersection as the dual-active set.
7. Extract scaffolds, apply the dual-pharmacophore SMARTS filter, enumerate analogues, and filter on synthetic accessibility.
8. Counter-screen against ACE and DPP-8/9.
9. If applying machine learning, follow Section 15, and use experimental labels with scaffold-based splitting.

18.3 Compute

A full-scale campaign of the order of 10^5 compounds per target requires several thousand CPU-hours and is intended for cluster execution. Every stage accepts a limit argument that truncates the working set, allowing the complete workflow to be exercised on a laptop before committing to a full run. Running the reduced version first is strongly advised; errors in grid definition or receptor preparation are considerably cheaper to discover at that scale.

19 Concluding remarks

The scientific contribution of this work is a reproducible protocol rather than a compound. Its more useful lesson is methodological.

Two of the defects identified here, the absence of enrichment benchmarking and the use of docking scores as machine learning labels, are not unusual. They appear regularly in published virtual screening studies. Both produce work that looks entirely normal and is substantially less informative than it appears, which is precisely what makes them worth naming.

The pharmacophore conflict at the centre of this project is genuine and unresolved. A molecule that inhibits both neprilysin and DPP-IV must carry opposing ionisable groups, and will pay for that in permeability. The pipeline described here is capable of finding such molecules. Whether any of them can be developed into a drug is a question that computation is not equipped to answer, and that only synthesis and assay can settle.

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Bibliographic details should be verified against the primary sources before formal citation. Software tools invoked by the pipeline carry their own licensing and citation requirements, listed in the accompanying repository.